

Description of the FOPPI-Controlled PV-RT Power System Cyberattack Dataset

This dataset is shared to support artificial intelligence-based studies aimed at distinguishing between normal and cyberattack conditions in PV-integrated two-area load frequency control (LFC) systems. It provides a reproducible data source for developing, testing, and comparing machine learning and deep learning-based cyberattack detection methods, including normal/attack discrimination and FDI/DoS attack classification.

The dataset contains simulation results obtained from the PV-RT power system controlled by the proposed Fractional-Order Predictive PI (FOPPI) controller. The PV-RT system consists of a photovoltaic (PV) generation unit in Area 1 and a reheat thermal generation unit in Area 2. The recorded variables represent the dynamic behavior of the controlled LFC system under random load changes and cyberattack scenarios.

The dataset includes both normal non-cyberattack scenarios and cyberattack scenarios involving false data injection (FDI) and denial-of-service (DoS) attacks. Each scenario is simulated for 15 seconds with a sampling time of 0.1 seconds; therefore, each scenario contains 151 time samples. Rows with the same ID value belong to the same scenario.

In the full dataset, 1440 scenarios are included in total: 720 normal scenarios and 720 cyberattack scenarios. Accordingly, the full time-domain dataset contains 217,440 rows. The scenario-based data can be used for deep learning model training, anomaly detection, statistical analysis, and controller robustness evaluation.

General Structure of the Dataset

Each row in the dataset represents one time sample of one scenario. The main columns are summarized below.

Column	Description
ID	Scenario number. The same ID value is used for all time samples belonging to the same scenario.
Area1-FDI / Area2-FDI	Binary indicators showing whether an FDI attack is applied to Area 1 and/or Area 2.
Area1-DoS / Area2-DoS	Binary indicators showing whether a DoS attack is applied to Area 1 and/or Area 2.
SLP1 / SLP2	Load perturbation or step load change values used in the scenario. In the provided benchmark, SLP2 represents the load change applied to the second area; SLP1 is included for area-wise scenario definition.
FDI1 / FDI2	False Data Injection attack coefficients applied to the ACE signals of Area 1 and Area 2. A value of 1 indicates no FDI attack; values greater than 1 represent the attack intensity.
DoS1 / DoS2	Denial-of-Service attack delay values applied to the ACE signals of Area 1 and Area 2. A value

	of 0 indicates no DoS attack; positive values represent the delay intensity in seconds.
ITAE	Integral of Time-weighted Absolute Error. It is recorded as a cumulative time-series performance index during the simulation.
ISE	Integral of Squared Error. It is recorded as a cumulative time-series performance index during the simulation.
ITSE	Integral of Time-weighted Squared Error. It is recorded as a cumulative time-series performance index during the simulation.
IAE	Integral of Absolute Error. It is recorded as a cumulative time-series performance index during the simulation.
df1	Frequency deviation of Area 1 in per unit.
df2	Frequency deviation of Area 2 in per unit.
dPtie	Tie-line power deviation between the two areas in per unit.
Time	Simulation time in seconds. The sampling time is 0.1 seconds.
Label	Class label of the scenario. Label 0 represents normal operation, Label 1 represents FDI attack, and Label 2 represents DoS attack.

Class Labels

The Label column indicates the class of each scenario. Random load variation is not considered a cyberattack by itself; therefore, normal scenarios may also include random load changes. In cyberattack scenarios, FDI or DoS effects are applied in addition to the load change condition.

Label	Description
0	Normal scenario without cyberattack
1	FDI attack scenario
2	DoS attack scenario

Scenario and Performance Interpretation

For FDI attacks, the ACE signal is proportionally distorted using the FDI1 and/or FDI2 coefficients. When the coefficient is equal to 1, no FDI attack is applied. Larger coefficients indicate stronger manipulation of the ACE signal.

For DoS attacks, the ACE signal is delayed before reaching the controller. The DoS1 and DoS2 columns define the delay values applied to the first and second area control loops, respectively.

The performance indices ITAE, ISE, ITSE, and IAE are cumulative signals calculated over time. For scenario-level controller performance evaluation, the final value or maximum value of these cumulative signals can be used. A lower ITAE value generally indicates better suppression of frequency and tie-line power deviations under the corresponding scenario.